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Unitedworld Institute of Technology (UIT)

Academic Year: - 2026-2027

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BRANCH:	CSE CORE
COURSE CODE:	26UTU01003
COURSE NAME:	FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE (FAI)



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Practical:1				
Sr. No.	Practical Name & Sub-Tasks	Tool Platform /	Mapped CO(s)	No. of Hours
1	Data Analysis to Dashboard. Sub-Tasks: - Select a small dataset and identify relevant attributes and analysis questions. - Clean and organize the dataset using an AI-assisted data analysis tool. - Create suitable charts and visualizations to represent key findings. - Develop a simple dashboard presenting visualizations, insights, and conclusions.	Gemini, Google Sheets /Excel, Looker Studio	CO2	2

• **Solution:**

1. Select a small dataset and identify relevant attributes and analysis questions.
Procedure: - Open Google Sheets / Microsoft Excel. - Create or import a small dataset such as Student Performance Data. - Add columns such as Student Name, Gender, Attendance, Maths, Science, and English Marks. - Enter the required records in the sheet. - Observe the dataset and identify important attributes.



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- Decide the questions to be analyzed, such as:
 - Which student scored the highest marks?
 - What is the average score of each student?
 - Which subject has the highest average?
 - Does attendance affect student performance?
- Save the dataset for further analysis.

Key Features Observed:

- Easy data entry and organization.
- Data can be arranged in rows and columns.
- Relevant attributes can be easily identified.
- Analysis questions can be created based on the available data.
- **Result:**
A suitable dataset was selected and the important attributes and analysis questions were successfully identified.

• Output:

	A	B	C	D	E	F	G	H	I	J	K
1	Student	Gender	Attendance %	Maths	Science	English	Average Marks				
2	Aarav	Male	92	85	78	82	81.66666667				
3	Diya	Female	88	76	91	87	84.66666667				
4	Rohan	Male	75	65	70	68	67.66666667				
5	Priya	Female	95	92	89	94	91.66666667				
6	Kabir	Male	68	58	62	65	61.66666667				
7	Ananya	Female	90	88	85	91	88				
8	Arjun	Male	80	72	75	70	72.33333333				
9	Meera	Female	85	81	84	86	83.66666667				
10	Vivaan	Male	72	67	64	69	66.66666667				
11	Isha	Female	93	90	92	88	90				



2. Clean and organize the dataset using an AI-assisted data analysis tool

Procedure:

- Open **Gemini** in a web browser.
- Sign in using a Google account.
- Upload the dataset or copy and paste the data into the chat.
- Enter a prompt such as:

“Check this dataset for missing values, duplicate records, incorrect values and suggest how to clean it.”

- Read the suggestions given by Gemini.
- Return to Google Sheets/Excel.
- Remove duplicate entries.
- Correct incorrect values or spellings.
- Fill missing data wherever required.
- Make sure all headings and number formats are correct.
- Save the cleaned dataset.

Key Features Observed:

- AI can quickly identify data errors.
- Detects missing and duplicate values.
- Suggests ways to clean and organize data.
- Saves time compared to manual checking.

Result:

- The dataset was successfully cleaned and organized using AI assistance and was ready for further analysis.



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• Output:

gemini.google.com/app/728cab329fd703ca

Analyze this student dataset. Find missing values, duplicate records, incorrect entries and suggest how to clean the data.

Show code

Based on the provided dataset from the Google Sheet, here is the dataset audit and cleaning breakdown:

Dataset Analysis Findings

- Missing Values: None. All 10 student entries (Rows 2-11) contain complete data across all columns.
- Duplicate Records: None. All student names, marks, and attendance values represent distinct records.
- Incorrect Entries: No mathematical errors found. The formula `=AVERAGE(D11:F11)` in column G accurately calculates the mean of Maths, Science, and English for each student.

Identified Data Quality Issues & Cleaning Recommendations

- Inconsistent Number Formatting
 - Issue: The `Average Marks` column displays raw, unformatted floating-point numbers with up to 8 decimal places (e.g., 81.6666667, 72.3333333), while exact averages display as whole integers (88, 90).
 - Fix: Select Column G and format the cells as Number with a fixed limit of 2 decimal

Show code

Upgrade

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 - Issue: The `Average Marks` column displays raw, unformatted floating-point numbers with up to 8 decimal places (e.g., 81.6666667, 72.3333333), while exact averages display as whole integers (88, 90).
 - Fix: Select Column G and format the cells as Number with a fixed limit of 2 decimal places (e.g., 81.67).
- Header Naming & Formatting
 - Issue: `Attendance %` includes a special character (%), which can cause issues when exporting data to programming scripts (Python, SQL, R).
 - Fix: Rename `Attendance %` to `Attendance_Percentage` or `Attendance_Pct`.
- Data Validation Setup (Preventative Cleaning)
 - Fix: Apply Data Validation rules to columns C through F to restrict numerical entries between 0 and 100 to prevent future entry errors.

Ask Gemini

Flash

Google - 138



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3. Create suitable charts and visualizations to represent key findings

Procedure:

- Open the cleaned dataset in Google Sheets/Excel.
- Select the required data.
- Click **Insert** → **Chart**.
- Create a **Bar/Column Chart** for Student vs Average Marks.
- Create a **Pie Chart** for category or gender distribution if required.
- Create a **Scatter Chart** for Attendance vs Marks.
- Add suitable chart titles and labels.
- Resize and arrange the charts properly.



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- Observe the charts and identify the main findings.

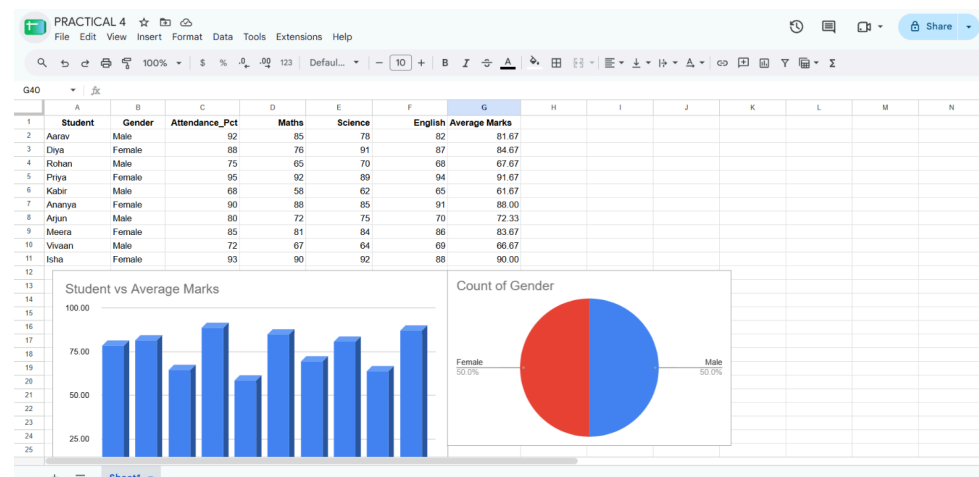
Key Features Observed:

- Different types of charts are available.
- Data can be converted into visual form easily.
- Charts make comparison easier.
- Trends and patterns can be understood quickly.
- Chart titles and labels can be customized.

Result:

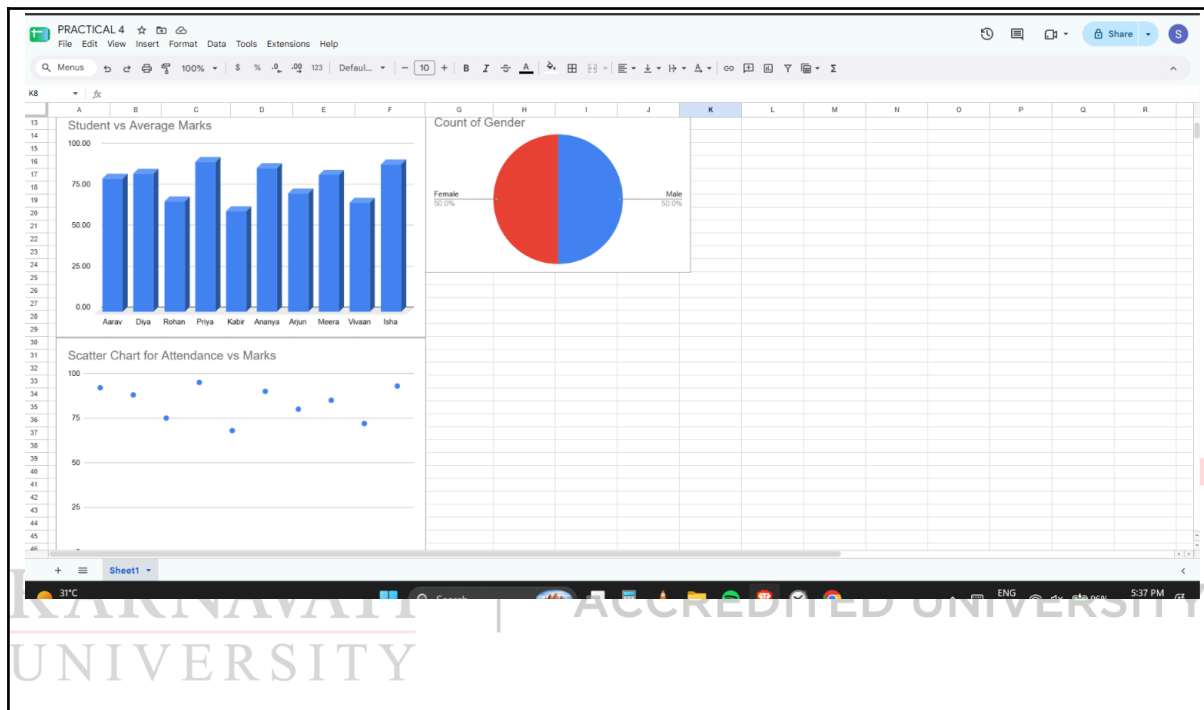
- Suitable charts and visualizations were successfully created to represent the important findings of the dataset.

Output:





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4. Develop a simple dashboard presenting visualizations, insights and conclusions.

Procedure:

- Create a new sheet and name it **Dashboard**, or open **Looker Studio**.



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- Add a title such as **Student Performance Dashboard**.
- Insert important values such as Total Students, Average Attendance and Average Marks.
- Add the charts created in the previous task.
- Arrange all charts and values properly on one page.
- Add short insights based on the analysis.
- Write a brief conclusion below the dashboard.
- Format the dashboard to make it clear and easy to understand.
- Save the final dashboard.

Key Features Observed:

- Multiple charts can be displayed in one place.
- Important values can be shown using scorecards or summary boxes.
- Makes analysis easy to understand.
- Provides a clear visual summary of the dataset.
- Helps present findings and conclusions effectively.

Result:

- A simple dashboard was successfully developed containing visualizations, important insights and conclusions from the analyzed dataset.

• Output:



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